Visit Alignment Is Learned but Nonmonotonic in Small Weight-Tied Residual Loops

Hybrid LDT/TRM Research Gym

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Abstract

Repeated use of one parameterized block changes how its update is accumulated and read through an unrolled computation. DeepLoop formalizes this effect with a visit-alignment coefficient κ_R, but leaves direct measurement as future work [1]. We estimate visit alignment from per-visit suffix sensitivities and single-live-application parameter gradients. At fixed loop counts, alignment is learned: a four-point descriptive exponent rises from 0.130 at initialization to 0.309 after 4,096 state-visit exposures. Across larger loop counts, however, the coefficient is not a stable power law. An initial terminal fit on $R \in \{2, 4, 8\}$ had $\gamma = 0.423$ and $R^2 = 0.994$; adding $R = 16$ reduced the same-horizon fit to 0.309. A sealed addendum then measured tied geometric-mean kappa 2.307 at $R = 32$ and 1.510 at untouched $R = 64$. The holdout missed both a sealed saturating-exponential prediction (2.551) and logarithmic prediction (2.877), returning the registered classification `form_unresolved`. The high-loop decline coincides with tied maximum gradient norms of 149–166, versus at most 1.565 in untied controls, while losses remain finite and decrease. A mini causal attention+MLP loop independently yields tied $\gamma = 0.501$ versus untied 0.018. These results reject $\gamma = 1$ as a descriptive constant in this small-scale regime, not as a conservative envelope. They do not establish asymptotic saturation, language-model performance, or a general stability threshold.

1 Motivation

A looped model applies a physical block repeatedly without increasing stored parameters. DeepLoop identifies the resulting tied-depth term: one shared update aggregates contributions from repeated visits and is then read through those visits in the next linearized forward pass [1]. Its analysis is controlled by κ_R, with decorrelated and fully aligned envelopes. The paper explicitly calls for direct measurement of κ_R or cross-round gradient alignment.

We ask four narrow questions. Does a direct estimator distinguish tied from untied visits? Is alignment fixed by architecture or learned during training? Does a small attention+MLP block reproduce the tied/untied contrast? Finally, does the measured dependence on R transfer beyond the original three loop counts? We make no task-accuracy or large-scale transfer claim.

2 Estimator and Schedule Correction

For each gradient-visible visit r, let U_r be a top-direction Jacobian vector product through the schedule suffix after that visit. Let G_r be the parameter gradient from a recomputed loss in which only visit r applies its module with live parameters; all other applications use detached parameters. We measure

$$\kappa_g = \frac{\|\sum_r U_r\| \, \|\sum_r G_r\|}{R_g \max_r \|U_r\| \max_r \|G_r\|}, \qquad 0 \le \kappa_g \le R_g. \tag{1}$$

Residual scale is excluded from κ_g and applied once in the schedule functional

$$B(A) = \sum_j J_j R_g(j) \kappa_g(j) \left(\frac{\beta_j}{\alpha_j}\right)^2. \tag{2}$$

Here $R_g(j)$ counts gradient-visible applications rather than all forward visits. This distinction is required for truncated or masked schedules; using $R(j)$ silently restores contributions that autograd cannot transmit.

For descriptive diagnostics we fit

$$\log \kappa_R = c + \gamma \log R \tag{3}$$

to geometric-mean kappas across three seeds. The estimator uses five power iterations and one fixed probe direction per seed at all checkpoints. The high-loop addendum instead preregisters two curves on raw kappa:

$$\kappa_{\text{sat}}(R) = K - Ae^{-R/\tau}, \qquad A \geq 0, \tag{4}$$

$$\kappa_{\log}(R) = a + b \log R, \qquad b \geq 0. \tag{5}$$

3 Registered Design

The primary model is a hidden-size-128 two-linear-layer residual loop on deterministic signed-permutation regression. Tied and visit-untied schedules use seeds $\{101, 103, 107\}$ and 4,096 state-visit exposures. Tied kappa is measured at exposures $\{0, 512, 1024, 2048, 4096\}$. The v0 three-point fit is frozen before observing $R = 16$:

$$\log \widehat{\kappa}_R = 0.025971 + 0.422909 \log R, \qquad \widehat{\kappa}_{16} = 3.315221. \tag{6}$$

The external cell replaces the block and task with a hidden-size-64 manual four-head causal attention+MLP residual block on deterministic prefix-context regression. Its strict gate requires both tied and untied fits to have $R^2 \geq 0.8$ and a gamma contrast of at least 0.1.

After the $R = 16$ result, a labeled addendum registers $R = 32$ as the last fit point and $R = 64$ as an untouched holdout. The two predictions must be committed and present on the remote branch before the holdout runner can construct an $R = 64$ cell. A model wins only with at least 0.05 absolute-error advantage and a winner-to-loser error ratio no greater than 0.75. Calling saturation also requires fit $R^2 \geq 0.9$ and predeclared tail conditions.

The stability ladder crosses $R \in \{6, 12\}$, $p \in \{0, 0.05, 0.10, 0.15\}$, learning rates $\{0.0003, 0.001, 0.003, 0.01\}$, and three seeds. A run is stable if finite, with maximum gradient norm at most 100 and final-to-initial loss ratio at most 10. Grid edges remain censored.

4 Alignment Grows During Training

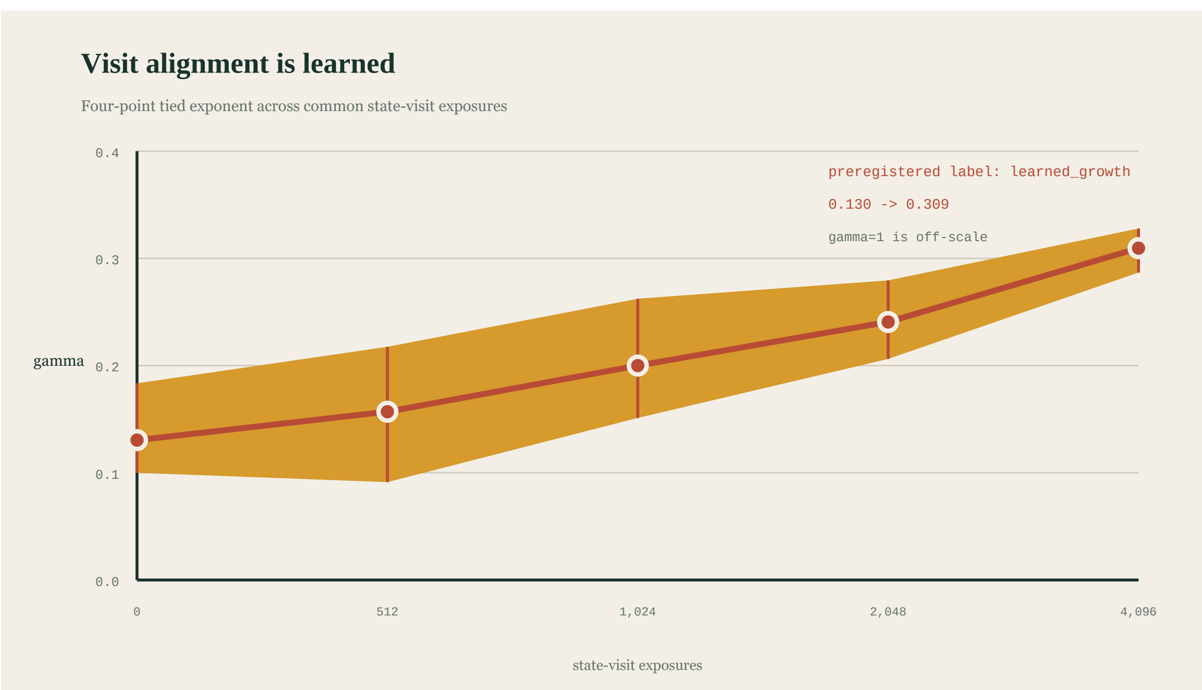

Figure 1: The four-point tied exponent and seed-bootstrap interval over common state-visit exposures. The registered outcome is `learned_growth`.

Table 1: Tied alignment exponent over training.

Exposures	γ	R^2	95% interval
0	0.1304	0.6116	$[0.1000, 0.1834]$
512	0.1570	0.6829	$[0.0913, 0.2176]$
1,024	0.2000	0.8214	$[0.1511, 0.2624]$
2,048	0.2406	0.8564	$[0.2062, 0.2794]$
4,096	0.3095	0.9140	$[0.2868, 0.3278]$

Gamma rises by 0.1790 and all four transitions are nondecreasing, satisfying the frozen `learned_growth` rule. Early R^2 values are weak. Thus the mechanism is more precise than "gamma increases": cross-visit alignment and its coherence across these four loop counts accumulate during training.

The fixed-direction terminal probes at $R = 2, 4, 8$ reproduce the sealed v0 kappas exactly. The held-out tied value is $\kappa_{16} = 2.550934$, a ratio of 0.769461 to the frozen prediction. Refitting all four points gives $\gamma = 0.309485$ and $R^2 = 0.914049$.

The v0 value 0.422909 and v0.1 value 0.309485 are not different-time measurements. They use the same 4,096-exposure horizon, hidden size, batch size, learning rate, residual scaling, and probe. The former fits $R = 2, 4, 8$; the latter refits the identical three observations after adding $R = 16$. The discrepancy is a support-set residual, not a failed replication.

5 Attention+MLP Replication

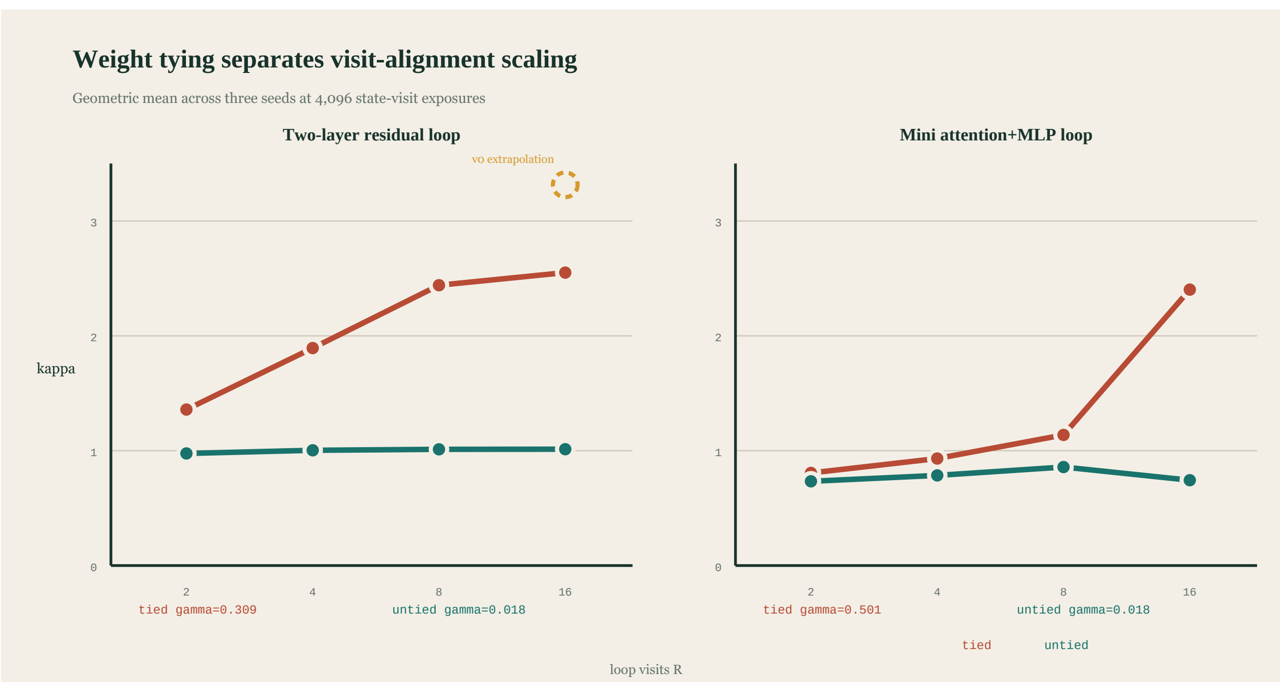

Figure 2: Terminal kappa in the primary residual MLP and mini attention+MLP cell. The open marker is the frozen v0 $R = 16$ extrapolation.

The mini attention+MLP loop gives tied $\gamma = 0.501246$ with $R^2 = 0.854021$. Its untied point estimate is 0.017859, but $R^2 = 0.050919$ because the four kappas form a noisy flat sequence rather than a power law. The gamma contrast is 0.483387.

The composite replication gate fails as registered because it requires a high-quality power-law fit from the intended flat negative control. This is a gate misspecification, not an evidence failure. A labeled post-hoc 20,000-draw seed bootstrap gives an untied gamma interval $[-0.092064, 0.077037]$, containing zero, with point magnitude below 0.1. This diagnoses the gate but does not retroactively convert it into preregistered confirmation. Future null controls should use a predeclared equivalence interval rather than power-law goodness of fit.

6 The High-Loop Holdout Rejects Both Smooth Forms

Figure 3: Tied kappa peaks at $R = 16$ and then falls while tied gradient scale rises sharply. Open markers are the two remotely sealed $R = 64$ predictions.

Table 2: Terminal alignment across loop count. Local gamma compares adjacent doubled loop counts.

R	tied κ	untied κ	local tied γ
2	1.357966	0.975547	–
4	1.893620	1.003657	0.479699
8	2.440651	1.011517	0.366119
16	2.550934	1.013226	0.063760
32	2.307170	1.005582	−0.144901
64	1.509898	1.097438	−0.611674

The saturating fit on $R \leq 32$ predicts $\kappa_{64} = 2.551038$; the logarithmic fit predicts 2.876785. Their absolute errors are 1.041141 and 1.366887. The saturating curve is closer, but its winner-to-loser error ratio 0.761687 misses the registered 0.75 limit. It also misses the fit-quality and monotone tail gates. The registered classification is therefore `form_unresolved`, not saturation.

All tied $R = 64$ cells remain finite and reduce loss, but their maximum gradient norms are 149.466, 165.824, and 160.771. Matched untied cells remain at or below 1.565. The coincidence makes a depth-by-training-state transition a live hypothesis: alignment may fall because trained high-loop dynamics enter a different regime, not because a stationary ceiling has been identified. The next measurement should map $\kappa(R, t)$ with denser checkpoints at $R \in \{16, 32, 64\}$.

Across all six loop counts, the untied point exponent is only 0.024572, but its narrow bootstrap interval $[0.018400, 0.033191]$ excludes exact zero. The addendum's registered flatness conjunction therefore also fails. This is reported without replacing the gate post hoc; future controls should test practical equivalence to zero.

7 The Stability Boundary Remains Censored

Figure 4: All aggregate cells through $R = 12$ remain stable. Every boundary is left-censored at $p \leq 0$, so no registered depth ordering is identifiable.

All 96 seed-level ladder runs are stable. The maximum observed gradient norm is 66.0719 and the maximum final-to-initial loss ratio is 0.6855. Every aggregate boundary is left-censored at $p \leq 0$ for both loop counts and all learning rates. This and the $R = 64$ stress are not inconsistent: the ladder stops at $R = 12$, while the addendum exposes a different high-loop regime.

8 Discussion

The direct measurements support five conclusions. First, weight tying creates measurable visit alignment in both tested model families. Second, alignment is partly learned rather than purely architectural. Third, the original $\gamma = 0.423$ fit is local rather than transferable. Fourth, neither registered smooth high-loop form predicts the untouched $R = 64$ result. Fifth, the fully aligned $\gamma = 1$ envelope is conservative rather than descriptive in this measured regime. We reject it as a measured constant, not as a sufficient bound.

If κ were eventually bounded, its local effective exponent would tend to zero and the associated DeepLoop threshold expression would tend toward $p = 1/4$. The addendum does not establish that premise. Even under bounded κ, the explicit R_g factor remains in $B(A)$, so total depth-dependent stability cost does not automatically vanish. The $R = 64$ gradient traces make this distinction empirical rather than merely algebraic.

To our knowledge, this is the first direct estimate of DeepLoop's specific visit-alignment coefficient, but that novelty statement is scoped to our recorded literature audit rather than an exhaustive survey.

9 Limitations and Reproducibility

The primary network is small and synthetic. The attention+MLP cell changes model and task together and is not a language model. Three seeds expose large high-loop variance but cannot characterize its distribution. The $R = 64$ gradient norms are diagnostics, not a preregistered divergence endpoint. No task-accuracy, sample-efficiency, asymptotic-saturation, large-scale Transformer, or general-stability claim follows.

Artifact hashes are listed below; each value is SHA-256.

- LSA v0.1 config:
  `c6cd78a09909956962a2c6d585270807088a9b8b0dc5e1e69fc8b151d45882cc`
- LSA v0.1 receipt:
  `3fa8c1e6f360035a8d1fc17932d18b2c6fe0249e6653c6e5aa9992a5cdf57414`
- saturation-addendum config:
  `f4eb1cd7d1df7a18f523498ed48a159ccc2f123c80957913301c51cc1cc30894`
- saturation-addendum receipt:
  `425e97fde47eb40b5baac381118ae95c77531b09083d2f847ee689951860b557`
- sealed $R = 64$ prediction at commit `d91b745`:
  `910f4d994be6d24afb8bb370b9b6e42a2b907df7b4b82245516286295119ed42`

Valid records comprise 42 replay, 72 primary, 24 attention+MLP, 96 ladder, six $R = 32$, and six untouched $R = 64$ observations. The addendum used 318.354 seconds of its one-hour hard-capped budget; all resource and cleanup receipts pass.

References

[1] Shuzhen Li, Yifan Zhang, Jiacheng Guo, Quanquan Gu, and Mengdi Wang. DeepLoop: Depth scaling for looped Transformers, 2026.